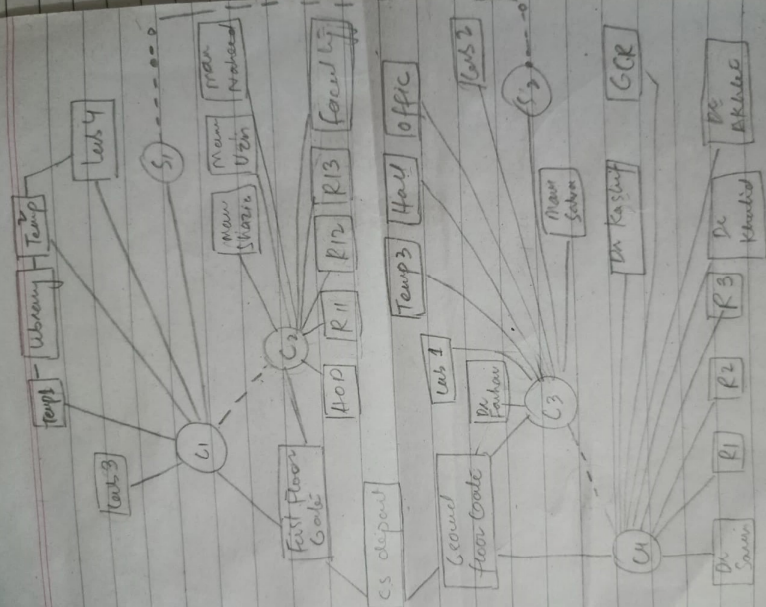


CS Department Map



Class 23/12/29, 23/12/110

Sent No: BSES, Sec 6A, Section B

Artificial Intelligence

Problem: Go to hall from library.

Q. Formulate the problem

A (i) Initial State: Library

(ii) Goal State: Hall

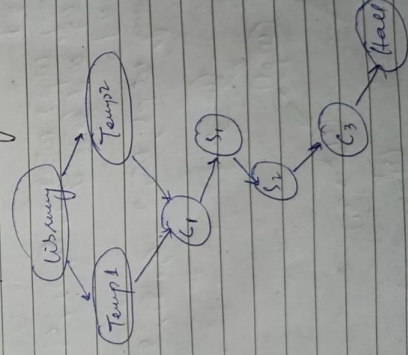
(iii) State space: CS department

(iv) Action: Move from one state to another

(v) Transition Model: To search from library to hall.

(vi) Cost-Path: Distance between nodes $d=1$

Q. Find path from library to hall.



- CS dept $\rightarrow$ Ground floor gate, first floor gate
- first floor gate $\rightarrow$ C1, C2
- C1 $\rightarrow$ Lab3, Temp1, Temp2, Lab4, S1, C2
- Temp 2 $\rightarrow$ Library
- Library $\rightarrow$ Temp2
- S1 $\rightarrow$ S2
- C2 $\rightarrow$ HOD, R11, R12, R13, Faculty
- Ground floor gate $\rightarrow$ C4, C3
- C3 $\rightarrow$ C4, Dr. Farhan, Lab1, Temp3, Hall, office, Lab2, S2, Main Salun
- C4 $\rightarrow$ Dr. Salim, R1, R2, R3, Dr. Khalid, Dr. Akhter

Date:

Q. Find the shortest path

A) Initialization:

C1: $Q_1 = \text{Library}$

$V_1 = \text{Library}$

(ii) deque library and expand

$Q_2 = \text{Temp1, Temp2}$

$V_2 = \text{Library, Temp1, Temp2}$

(iii) deque Temp1 and expand

$Q_3 = \text{Temp2, C1}$

$V_3 = \text{Library, Temp1, Temp2, C1}$

(iv) deque Temp2 and expand

$Q_4 = \text{S1}$

$V_4 = \text{Library, Temp1, Temp2, C1, S1}$

(v) deque S1 and expand

$Q_5 = \text{S2}$

$V_5 = \text{Library, Temp1, Temp2, C1, S1, S2}$

(vi) deque S2 and expand

$Q_6 = \text{C3}$

$V_6 = \text{Library, Temp1, Temp2, C1, S1, S2, C3}$

(vii) deque C3 and expand

$Q_7 = \text{Hall}$

$V_7 = \text{Library, Temp1, Temp2, C1, S1, S2, C3, Hall}$

(viii) deque Hall and expand

$Q_8 = \text{Hall}$

$V_8 = \text{Library, Temp1, Temp2, C1, S1, S2, C3, Hall, Hall}$

(ix) deque Hall and expand

$Q_9 = \text{Hall}$

$V_9 = \text{Library, Temp1, Temp2, C1, S1, S2, C3, Hall, Hall, Hall}$

(x) deque Hall and expand

$Q_{10} = \text{Hall}$

$V_{10} = \text{Library, Temp1, Temp2, C1, S1, S2, C3, Hall, Hall, Hall, Hall}$

(xi) deque Hall and expand

$Q_{11} = \text{Hall}$

$V_{11} = \text{Library, Temp1, Temp2, C1, S1, S2, C3, Hall, Hall, Hall, Hall, Hall}$

(xii) deque Hall and expand

$Q_{12} = \text{Hall}$

$V_{12} = \text{Library, Temp1, Temp2, C1, S1, S2, C3, Hall, Hall, Hall, Hall, Hall, Hall}$